

CSCI212 : Lecture #14

Kevin Bedoya

Wednesday 3/23/26

Coding GUIs & Pictures

GUI = Graphical User Interface

It is a device that allows you to interact with the computer graphically. In Java, we will use **Swing** (a classic Java GUI toolkit) a modern alternative is **JavaFX** (but more challenging to setup, it is not native to the Java SDK). Java Swing is an *extension* of AWT (Abstract Window Toolkit).

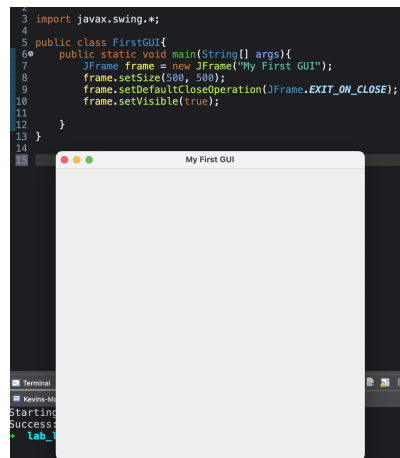
Java Swing grants us access to GUI components, e.g. **JFrame**, **JButton**, **JLabel**, ect.

GUI programming differs from the conventional style we learned in class, in particular, we will focus on organizing components on screen, and supplying functionality to components, e.g. clicking a button on screen prints a message to the console (this is an example of *event-driven programming*).

Example #1

Let us place a window onto screen. In order for this work, we need to be make use of **JFrame**; JFrame is a class that represents a window on screen. Before using JFrame, we will need to import Java Swing in our scope.

```
1 import javax.swing.*;
2
3 public class FirstGUI{
4     public static void main(String[] args){
5         JFrame frame = new JFrame("My First GUI");
6         frame.setSize(500, 500);
7         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
8         frame.setVisible(true);
9     }
10 }
```



JButton (Buttons)

We want buttons and other parts of the GUI to respond to users. We need to supply the GUI a function to perform an action in response to an *event* (e.g the user clicking a button), this type of function is called a *callback* function.

How can we add a callback function? We will code this with a lambda. In particular a *Functional Interface*, e.g. `ActionListener`. We provide a single method:

```
1 void actionPerformed(ActionEvent e);
```

In practice, we have:

```
1 JButton b = new JButton("Generic button");
2 b.addActionListener(e -> System.out.println("Button clicked!"));
3 // The lambda determines what action should follow upon execution of the event.
4 frame.add(button);
5 // Make sure to include the button onto the frame otherwise it will not appear!
```

In the example below, the button is clicked twice.

